

Tuning observation report

Experiment #35 · HIGH · COMPLETED

Generated: 2026-09-20T12:16:15.624Z · Started: 2026-09-20T08:17:39.707943+00:00 · Ended: 2026-09-20T08:22:37.835286+00:00

ROLLBACK

Degraded or insufficient improvement across observation metrics

Action: 23ab8515-edf1-4723-9ba5-ebd7312e47f5

Setting: max_parallel_workers_per_gather · 2 → 1

Execution: Automatic · Scope: bound_workload_group · Bound sessions: 10

Action timestamp: 2026-09-20T08:21:41.797242Z · Observed duration: 33.15 seconds

Verified applied value: 1 · Verified restored value: 2

Baseline samples: 3 · Observation samples: 6

Baseline window: 2026-09-20T08:21:31.751607+00:00 — 2026-09-20T08:21:41.797242+00:00

METRIC	BEFORE	AFTER	DIFFERENCE	CHANGE
Query latency ms	881.98	918.02	36.05	4.09%
Database throughput TPS (database-wide)	45.94	38.39	-7.55	-16.42%
CPU utilization %	99.73	99.88	0.15	0.15%
Memory utilization %	10	9.97	-0.03	-0.33%
Disk read bytes / interval	0	0	0	Unavailable
Disk write bytes / interval	102,400	70,997.33	-31,402.67	-30.67%

Negative latency change means faster responses. Positive throughput change means more database transactions. Resource changes are descriptive, not performance verdicts.

ROLLBACK

Degraded or insufficient improvement across observation metrics

Action: 02f085e3-a5c2-4852-b2d5-bc5bc241547a

Setting: max_parallel_workers_per_gather · 2 → 1

Execution: Automatic · Scope: bound_workload_group · Bound sessions: 10

Action timestamp: 2026-09-20T08:20:26.694229Z · Observed duration: 33.2 seconds

Verified applied value: 1 · Verified restored value: 2

Baseline samples: 3 · Observation samples: 6

Baseline window: 2026-09-20T08:20:16.702299+00:00 — 2026-09-20T08:20:26.694229+00:00

METRIC	BEFORE	AFTER	DIFFERENCE	CHANGE
Query latency ms	938.81	891.99	-46.83	-4.99%
Database throughput TPS (database-wide)	41.4	41.69	0.29	0.71%
CPU utilization %	99.67	99.88	0.22	0.22%
Memory utilization %	9.93	9.93	0	0%
Disk read bytes / interval	0	0	0	Unavailable
Disk write bytes / interval	98,304	81,920	-16,384	-16.67%

Negative latency change means faster responses. Positive throughput change means more database transactions. Resource changes are descriptive, not performance verdicts.

ROLLBACK

Degraded or insufficient improvement across observation metrics

Action: 8836d223-5712-4d3f-9265-1fb7f60addc3

Setting: max_parallel_workers_per_gather · 2 → 1

Execution: Automatic · Scope: bound_workload_group · Bound sessions: 10

Action timestamp: 2026-09-20T08:19:09.918848Z · Observed duration: 33.13 seconds

Verified applied value: 1 · Verified restored value: 2

Baseline samples: 3 · Observation samples: 6

Baseline window: 2026-09-20T08:18:59.907877+00:00 — 2026-09-20T08:19:09.918848+00:00

METRIC	BEFORE	AFTER	DIFFERENCE	CHANGE
Query latency ms	913.05	900.31	-12.73	-1.39%
Database throughput TPS (database-wide)	43.8	39.6	-4.2	-9.59%
CPU utilization %	99.7	99.87	0.17	0.17%
Memory utilization %	9.97	9.93	-0.03	-0.33%
Disk read bytes / interval	0	2,730.67	2,730.67	Unavailable
Disk write bytes / interval	73,728	91,477.33	17,749.33	24.07%

Negative latency change means faster responses. Positive throughput change means more database transactions. Resource changes are descriptive, not performance verdicts.

ROLLBACK

Degraded or insufficient improvement across observation metrics

Action: 8115cd78-fa89-44d1-bb1e-4b414e3e06a8

Setting: max_parallel_workers_per_gather · 2 → 1

Execution: Automatic · Scope: bound_workload_group · Bound sessions: 10

Action timestamp: 2026-09-20T08:17:53.258541Z · Observed duration: 33.22 seconds

Verified applied value: 1 · Verified restored value: 2

Baseline samples: 3 · Observation samples: 6

Baseline window: 2026-09-20T08:17:44.824216+00:00 — 2026-09-20T08:17:53.258541+00:00

METRIC	BEFORE	AFTER	DIFFERENCE	CHANGE
Query latency ms	813.85	827.4	13.54	1.66%
Database throughput TPS (database-wide)	51.41	42.14	-9.27	-18.03%
CPU utilization %	99.7	99.87	0.17	0.17%
Memory utilization %	9.93	9.92	-0.02	-0.17%
Disk read bytes / interval	0	0	0	Unavailable
Disk write bytes / interval	195,242.67	73,045.33	-122,197.33	-62.59%

Negative latency change means faster responses. Positive throughput change means more database transactions. Resource changes are descriptive, not performance verdicts.

Scope & limitations

A before/after observation does not establish causation or a general performance improvement. Throughput is database-wide TPS, not direct owned-workload queries per second. Other activity can affect telemetry.

OS settings were not automatically tuned. Missing values are unavailable, never zero-filled.

Use a completed paired evidence study for a stronger performance claim. KEEP alone is not proof.